

Image Segmentation Using Local Variation and Edge-Weighted Centroidal Voronoi Tessellations

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Abstract—The classic centroidal Voronoi tessellation (CVT) model and its generalizations work quite well at extracting uniformly colored objects, but often fail to handle images with distinct color distribution or strong inhomogeneous intensity. To resolve this problem within the CVT methodology, in this paper we incorporate the information of local variation of colors/intensities and the length of boundaries into the energy functional and develop a new model called the *Local Variation and Edge-Weighted Centroidal Voronoi Tessellation* (LVEWCVT) for image segmentation. Its mathematical formulation and practical implementations are also discussed and given. We test the LVEWCVT method on various type of segments and also compare it with several state-of-art algorithms using extensive segmentation examples, the results demonstrate excellent performance and competence of the proposed method.

Index Terms—Centroidal Voronoi, centroidal Voronoi tessellations (CVT), clustering, edge-weighted, image segmentation, intensity inhomogeneity, intensity variation.

I. INTRODUCTION

IMAGE segmentation is essentially the process of dividing an image into small pieces or objects that have a common attribute, e.g., color, brightness, texture, boundary continuity and so on. Moreover, the resulting segments must be “visually meaningful” [2] although it is quite hard to give “visually meaningful” a precise and quantitative definition. Different people may like to divide the same image in different ways according to their practical purposes. Furthermore, huge diversity of images in the real world are getting more and more complicated with advances of photography techniques and painting tools. Therefore, image segmentation is still a very difficult problem.

Traditionally, there are two main approaches to address this challenging task: the *edge-based* and *region-based* approaches. The idea of edge-based segmentation is to detect the locations of discontinuities in brightness, color, texture etc. However, the resulting image by edge detection is essentially “not” a segmentation result yet [3], [32], [40], because the edge fragments need to

be linked into closed boundaries that correspond to meaningful objects in the further processing stages. In contrast, the region-based algorithms group the pixels with similarities into subregions directly, e.g., the K-means clustering and GMM based approaches [5], [16], [20], [21], thus the boundaries can be easily obtained from the constructed subregions. The basic idea behind region-based approaches is to segment the images into regions with maximum homogeneity. Various criteria for homogeneity can be utilized such as intensity, color, texture, shape, model (using semantic information) and so on [40].

Many recent works attempted to combine both of the region and edge cues [31], [33], [36], [37], [38], [39] and have lead to great success. In the original graph-based algorithms [18], [39], similarities and dissimilarities are measured between each pair of pixels to determine if there is an intervening contour between them. This approach has been successfully applied to a large set of natural image datasets, but this approach is usually computationally expensive [39]. The method of segmentation by weighted aggregation (SWA) [38], a split-and-merge method, has largely improved the accuracy and efficiency of this type of algorithms, but it is usually not robust enough to noise distortion [17]. Moreover, in order to generate a segmentation which is accurate enough, the methods like normalized cuts [39] and SWA often have relatively high labeling costs, i.e., large number of fragments to cover the objects in the foreground [17]. The level set method is another popular approach which extracts objects using front propagation [6], [7], [25], [28], [43], [44]. Main advantages of level set methods include that the objects topology can be changed, they are almost parameter free, and any type of energy can be used. However, for multi-phase image segmentation, the level set methods need substantially more efforts [19], [44].

Therefore, it is still highly desired to come up with a new algorithm which is robust to the noise, accurate yet with relatively low labeling cost, and computationally efficient. One of the possible candidates is the so-called *Centroidal Voronoi tessellations* (CVTs). CVTs are special Voronoi tessellations whose generators are also the centers of mass (centroids) of the Voronoi regions with respect to a given density function. Construction of CVTs is often led by the minimization of an energy functional. In its simplest and classic form, CVT-based algorithms reduce to well-known *k*-means clustering technique [12] (This is part of the reason that the CVT-based algorithms are usually computationally efficient). In the past decade, the CVT methodology has attracted a lot of attentions in many applications due to its generality and universality [4], [9], [10], [11], [13], [14].

Unfortunately, the classic CVT model is sensitive to noisy data, i.e., it may result in many small fragments in the process

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of segmenting images. By adding a new energy term related to the edge length to the original clustering energy functional, we developed in [45] a so-called *Edge-Weighted CVT* (EWCVT) model for image segmentation, which is pretty robust to noises and hence able to extract the objects more accurately. In the EWCVT model, pixels belonging to the same cluster share the same centroid and hence the segmentation results produced by the EWCVT method are still relatively homogeneous.

In this paper we propose a novel weighted CVT model and corresponding algorithms for image segmentation. Unlike the classic CVT in which all pixels inside the same cluster share a common centroid, we will assign each pixel a “*locally*” computed centroid. Therefore, even for pixels which belong to the same cluster, their “*centroids*” are not necessarily the same. Similar to the EWCVT model, the energy functional in the new model consists of two parts, one is the new “*local variation*” energy instead of the classic clustering energy, and the other is still the edge energy related to the boundary length. We will assign different clusters different weights when evaluating their local energy. This scheme enables our new model to capture the objects with sharp variations in intensity or color. Furthermore, the edge energy term in our model is very helpful in dealing with noises [45].

This paper is organized as follows. In Section II, we first give a brief review on the classic CVT and the improved version—EWCVT proposed in [45]. The new LVEWCVT model and its implementations are then developed in details in Section III. Applications of the LVEWCVT model and extensive numerical examples are presented in Section IV, together with some discussions and quantitative results to demonstrate efficiency, effectiveness, robustness, and flexibility of the proposed method. Some concluding remarks are finally given in Section V.

II. CVT AND IMAGE SEGMENTATION

In this section, we briefly summarize some basic concepts of CVTs and its variants for image segmentation. For details, please refer to [12], [45] and references cited therein.

Image segmentation is practically a data grouping problem. The data normally refers to pixels. Therefore an image can be regarded as a function f which maps a set of pixels $\mathcal{D}$ to gray levels or colors. Clearly, the set of pixels $\mathcal{D}$ is a discrete set. Normally, in the 2D or 3D case, pixels are usually denoted by their coordinates which are positive integers, thus elements in $\mathcal{D}$ are in fact integer pairs or triples. For simplicity, we use bold lowercase letters, like $\mathbf{p}, \mathbf{q}$, etc., to denote the elements (pixels) in $\mathcal{D}$.

A. CVT and Clustering Energies

CVTs are special Voronoi tessellations whose generators coincide with the corresponding centroids of the Voronoi regions, with respect to a given density function. Next we introduce some basic concepts and definitions precisely.

Definition 1: Let $\mathcal{I} = \{f(\mathbf{p})\}_{\mathbf{p} \in \mathcal{D}}$ denotes the image of f over $\mathcal{D}$ and $\mathcal{G} = \{g_l\}_{l=1}^L$ a set of typical colors (or brightness

levels). The *Voronoi region* (or *Voronoi cluster*) V_k in $\mathcal{D}$ associated with the color g_k is defined by

$$V_k = \{\mathbf{p} \in \mathcal{D} : |f(\mathbf{p}) - g_k| < |f(\mathbf{p}) - g_l|, \text{ for } l = 1, \dots, L, l \neq k\}, \quad k = 1, \dots, L. \quad (1)$$

where $|\cdot|$ is certain predefined metric such as the Euclidean one, g_k is the so-called *generator* of V_k . The set $\mathcal{V} = \{V_k\}_{k=1}^L$ is called a *Voronoi tessellation* of $\mathcal{D}$ [29], [12].

Note that according to definition 1, we have $V_i \cap V_j = \emptyset$ if $i \neq j$ and $\mathcal{D} = \cup_{k=1}^L V_k$. Therefore the Voronoi tessellation $\mathcal{V}$ is a special partition of $\mathcal{D}$.

Definition 2: Given a partition of $\mathcal{D}$, denoted by $\{S_k\}_{k=1}^L$, the *centroid* (*center of mass* or *cluster mean*) of each segment S_k is defined as

$$\bar{g}_k = \arg \min_g \sum_{\mathbf{p} \in S_k} |f(\mathbf{p}) - g|^2. \quad (2)$$

For an arbitrary Voronoi tessellation $(\{g_k\}_{k=1}^L; \{V_k\}_{k=1}^L)$ of $\mathcal{D}$, usually $g_k \neq \bar{g}_k$ for $k = 1, \dots, L$.

Definition 3: If the generators of a Voronoi tessellation $\{V_k\}_{k=1}^L$ of $\mathcal{D}$ coincide with its corresponding centroids, i.e.,

$$g_k = \bar{g}_k, \quad k = 1, \dots, L$$

then we call the Voronoi tessellation $\{V_k\}_{k=1}^L$ a CVT of $\mathcal{D}$ [12], [13] and refer to $\{g_k\}_{k=1}^L$ as the corresponding CVT generators.

When a metric is given, the corresponding CVTs often can be effectively constructed by the Lloyd’s algorithm [26]. On the other hand an interesting fact is that it has been rigorously proved that the construction of CVTs is equivalent to an energy minimization process, provided that a properly defined energy functional is consistent with the metric defined on the data sets [13]. In fact, we have the following theorem.

Theorem 1: For any set of generators $\mathcal{G} = \{g_k\}_{k=1}^L$ and any partition $\mathcal{S} = \{S_k\}_{k=1}^L$ of $\mathcal{D}$, let us define the *classic clustering energy* as follows:

$$E(\mathcal{G}; \mathcal{S}) = \sum_{k=1}^L \sum_{\mathbf{p} \in S_k} |f(\mathbf{p}) - g_k|^2. \quad (3)$$

Then $E(\mathcal{G}; \mathcal{S})$ is minimized only if $(\mathcal{G}; \mathcal{S})$ form a CVT of $\mathcal{D}$, i.e., $\mathcal{S}$ are Voronoi regions of $\mathcal{D}$ associated with the generators $\mathcal{G}$ and simultaneously $\mathcal{G}$ are the corresponding centroids of the regions $\mathcal{S}$.

For detailed proof and discussions of Theorem 1, please refer to [13] and [45].

B. The Classic CVT-Based Algorithms for Image Segmentation

Numerical implementations of the classic CVT model are pretty straightforward. There are lots of studies on how to construct the classic CVTs efficiently, e.g., [22], [41], [42], and [45]. In particular, the most popular Lloyd’s algorithm [26] can be summarized as follows:

- 1) given an arbitrary partition as starting point;
- 2) compute the centroids of all segments of the partition;

- 3) use the centroids as generators to compute the corresponding Voronoi regions (segments);
- 4) repeat steps 2 and 3 until the centroids and generators agree with each other.

In [45], two implementations were proposed, Algorithm CVT-A and Algorithm CVT-B, to compute the classic CVTs based on the Lloyd's iteration. In both Algorithm CVT-A and Algorithm CVT-B, the metric defined in the color space is adopted for computing the Voronoi regions, as illustrated by (1). To some extent Algorithm CVT-B is an accelerated and improved version of Algorithm CVT-A. For discussion on convergence and stability of the above implementations, we refer to [12], [15], and [23].

C. Deficiencies of the Classic CVT Model

The classic CVT model works well in extracting uniform colored regions, but is sensitive to noises [12], [45]. Fig. 1 provides two examples to illustrate the difficulties of classic CVTs. Images in the left column of Fig. 1 are the original images. The upper one is the "Europe-at-night" and the lower one is a medical image with a vessel. Segmentation results by the classic CVT method are presented by images in the middle column. These images have some typical intensity inhomogeneities: the "Europe-at-Night" image consists of many large light spots and is quite noisy inside the Europe landmass; on the other hand, the blood vessel image has a darker background in the bottom-left corner of the image and thus the local intensity variation in that part is smaller. For both images, the classic CVT model fails to provide satisfying segmentation results; in fact, the Europe landmass and the vessel can never be identified as a whole no matter how many clusters are used in the CVT model, as shown in the middle column.

D. EWCVT Model and Its Deficiencies

To overcome the difficulties of the classic CVT method, a generalized model, the EWCVT model, was proposed in [45], which combines a new energy term $E^{\mathcal{L}}$ (called *edge energy*) with the classic clustering energy such as

$$\hat{E}(\mathcal{G}, \mathcal{S}) = E(\mathcal{G}, \mathcal{S}) + E^{\mathcal{L}}(\mathcal{S}). \quad (4)$$

Naturally, we call the minimizer $(\tilde{\mathcal{G}}, \tilde{\mathcal{S}})$ of $\hat{E}(\mathcal{G}, \mathcal{S})$ an EWCVT.

To rigorously define the edge energy, let us give some notations first. For each pixel $\mathbf{p} \in \mathcal{D}$, $\mathbb{N}_r(\mathbf{p})$ represents a local neighborhood of $\mathbf{p}$, which is a disk centered at $\mathbf{p}$ with radius r . We call any pixel $\mathbf{q} \in \mathbb{N}_r(\mathbf{p})$ a neighbor pixel of $\mathbf{p}$. $\pi_{\mathcal{S}}(\cdot)$ is defined as a projection function: $\mathcal{D} \rightarrow \{1, 2, \dots, L\}$ which maps each pixel $\mathbf{p}$ to the index of the cluster that $\mathbf{p}$ belongs to.

For neighbor pixels $\mathbf{q}$ of $\mathbf{p}$, we define the characteristic function $\chi_{\mathbf{p}}: \mathbb{N}_r(\mathbf{p}) \rightarrow \{0, 1\}$ as

$$\chi_{\mathbf{p}}(\mathbf{q}) = \begin{cases} 1, & \text{if } \pi_{\mathcal{S}}(\mathbf{p}) \neq \pi_{\mathcal{S}}(\mathbf{q}); \\ 0, & \text{otherwise.} \end{cases} \quad (5)$$

Then the edge energy for each $\mathbf{p} \in \mathcal{D}$ can be written as

$$\mathcal{E}_{\mathcal{S}}^{\mathcal{L}}(\mathbf{p}) = \mu \sum_{\mathbf{q} \in \mathbb{N}_r(\mathbf{p})} \chi_{\mathbf{p}}(\mathbf{q}). \quad (6)$$

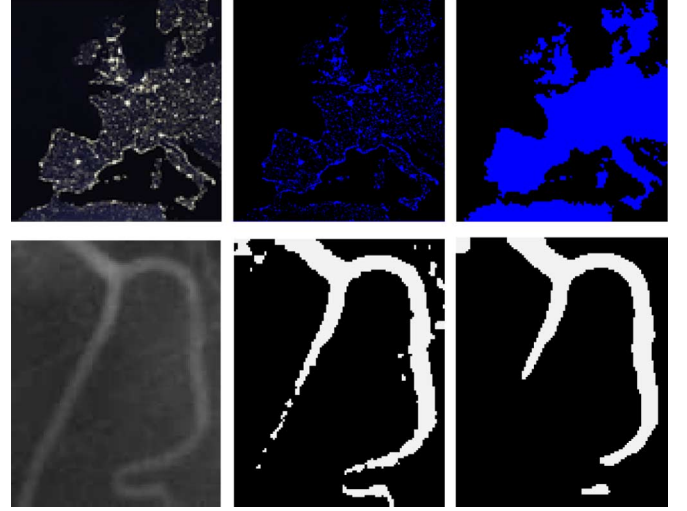


Fig. 1. Results of segmenting two images with some typical intensity inhomogeneities by the classic CVT and EWCVT models. (top row) Image of "Europe-at-Night." (bottom row) Image of blood vessel. (left column) Original images. (middle column) Segmentation results by the classic CVT model. (right column) Segmentation results by the EWCVT model.

where $\mu > 0$ is a constant weighting parameter. The subscript $\mathcal{S}$ indicates that the edge energy only depends on the current segmentations. For simplicity, we will omit the subscript from now on.

Naturally, the total edge energy is the summation of $\mathcal{E}^{\mathcal{L}}(\mathbf{p})$ over $\mathcal{D}$

$$E^{\mathcal{L}}(\mathcal{S}) = \sum_{\mathbf{p} \in \mathcal{D}} \mathcal{E}^{\mathcal{L}}(\mathbf{p}). \quad (7)$$

It was proved [45] that $E^{\mathcal{L}}(\mathcal{S})$ is "proportional to the true boundary length" as $r \rightarrow 0$. Therefore seeking the minimizer of $\hat{E}(\mathcal{G}, \mathcal{S})$ will naturally bound the boundary length. As a useful consequence, we can greatly reduce the noisy data in the form of small blocks by limiting the boundary length, i.e., the edge energy $E^{\mathcal{L}}$ [45]. In general, this scheme is a very effective approach to control the smoothness of the resulting boundaries (i.e., regularity of the segmentation).

The upper right image of Fig. 1 shows the segmentation result by the EWCVT model of the Europe-at-night image. We can see that the Europe landmass is accurately identified. However, since EWCVT's energy includes the classic clustering energy, pixels inside the same segment (cluster) still need to be roughly the same with respect to the centroid. Therefore, the EWCVT often fails to correctly identify the objects with huge intensity/color variations, e.g., the vessel image in Fig. 1. Although the intensities vary gradually within the vessel, they differ a lot for far away pixels. From the lower right image of Fig. 1, we can see that the EWCVT segmentation result loses a large portion of the vessel, although the boundary is pretty smooth.

III. LOCAL VARIATION AND EDGE-WEIGHTED CENTROIDAL VORONOI TESSELLATIONS (LVEWCVTs)

Deficiencies of the classic CVT and EWCVT models are mostly due to the definition of centroids. Since the centroid is the mean of a whole segment/cluster in certain way in these models,

there is no surprise that minimization of the classic clustering energy would lead to relatively uniform subregions. Therefore, completely different with the classic CVT and EWCVT, we conceive that the centroids could be defined and calculated locally for each pixel. Motivated by this idea, we develop a new *Local Variation and Edge-Weighted Centroidal Voronoi Tessellation* (LVEWCVT) model for image segmentation in the following.

In this section, we first present the mathematical formulation of the energy functional of the LVEWCVT model. Then we derive a metric which is proven to be consistent with the energy functional. As discussed in [45], once we have the metric defined, LVEWCVT can be easily constructed by the Lloyd type algorithm or its variations.

A. Local Variation Energy

The centroids defined in classic CVTs are a measure of global information to some extent. Due to shortcomings mentioned above, we need find a measure which is able to capture and describe the local intensity information around each pixel. We will call this measure the “*local variation energy*” of each pixel.

There is still one more issue we need to consider. Should every neighbor pixel $\mathbf{q}$ of $\mathbf{p}$ have almost same effect on the local variation energy of the pixel $\mathbf{p}$? Clearly it is more reasonable that far away neighbor pixels should have less effect than the nearby pixels. Therefore, we need a weight function $H : \mathcal{D} \times \mathcal{D} \rightarrow [0, +\infty)$ such that for any $\mathbf{p}, \mathbf{q} \in \mathcal{D}$:

- 1) $H(\mathbf{p}, \mathbf{q}) \geq 0$;
 - 2) $H(\cdot, \cdot)$ is symmetric, i.e., $H(\mathbf{p}, \mathbf{q}) = H(\mathbf{q}, \mathbf{p})$;
 - 3) $H(\mathbf{p}, \mathbf{q}_1) \geq H(\mathbf{p}, \mathbf{q}_2)$, if $|\mathbf{p} - \mathbf{q}_1| \leq |\mathbf{p} - \mathbf{q}_2|$, and $\lim_{|\mathbf{p}-\mathbf{q}|\rightarrow\infty} H(\mathbf{p}, \mathbf{q}) = 0$;
- in which $|\cdot|$ denotes the Euclidean distance.

Then the *local variation energy* for each pixel $\mathbf{p} \in \mathcal{D}$ is defined as:

$$\mathcal{E}_{(\tilde{\mathcal{G}}; \mathcal{S})}^{\text{LV}}(\mathbf{p}) = \sum_{k=1}^L \left(\lambda_k \sum_{\mathbf{q} \in \mathbb{N}_r(\mathbf{p}) \cap S_k} H(\mathbf{p}, \mathbf{q}) |f(\mathbf{q}) - \tilde{g}_k(\mathbf{p})|^2 \right) \quad (8)$$

where $\lambda_k, k = 1, 2, \dots, L$, are some positive constants. Note that different with the set of generators in the classic CVT model (a set of points), $\tilde{\mathcal{G}} = \{\tilde{g}_k(\cdot)\}_{k=1}^L$ are in fact a set of functions defined over the whole domain $\mathcal{D}$, thus we call $\tilde{\mathcal{G}} = \{\tilde{g}_k(\cdot)\}_{k=1}^L$ the *generator functions*. The subscript of $\mathcal{E}_{(\tilde{\mathcal{G}}; \mathcal{S})}^{\text{LV}}(\cdot)$ reminds us that the local variation energy depends on both the current segmentations and the generator functions. For simplicity, we will omit the subscript from now on. But please remember the local variation energy not only depends on the pixel but also on current configurations.

Moreover, it is almost straightforward that we can minimize (8) if we set

$$\tilde{g}_k(\mathbf{p}) = \frac{\sum_{\mathbf{q} \in \mathbb{N}_r(\mathbf{p}) \cap S_k} H(\mathbf{p}, \mathbf{q}) f(\mathbf{q})}{\sum_{\mathbf{q} \in \mathbb{N}_r(\mathbf{p}) \cap S_k} H(\mathbf{p}, \mathbf{q})}. \quad (9)$$

We also set

$$\tilde{g}_k^{\text{LBC}}(\mathbf{p}) = \arg \min_{\tilde{g}_k(\mathbf{p})} \mathcal{E}^{\text{LV}}(\mathbf{p})$$

which is called the local biased centroid (LBC) of the pixel $\mathbf{p}$ with respect to S_k .

Finally the local variation energy over the whole domain $\mathcal{D}$ is simply the summation of $\mathcal{E}^{\text{LV}}(\mathbf{p})$ over $\mathcal{D}$, i.e.,

$$E^{\text{LV}}(\tilde{\mathcal{G}}; \mathcal{S}) = \sum_{\mathbf{p} \in \mathcal{D}} \mathcal{E}^{\text{LV}}(\mathbf{p}). \quad (10)$$

Remark: There are many choices for the weight function H as long as it satisfies the three properties. In this paper, we choose the Gaussian kernel

$$H_\sigma(\mathbf{p}, \mathbf{q}) = \frac{1}{(2\pi)^{n/2} \sigma^n} e^{-|\mathbf{p}-\mathbf{q}|^2/2\sigma^2} \quad (11)$$

with a scale parameter $\sigma > 0$. Note the Gaussian kernel is very close to zero when $|\mathbf{p} - \mathbf{q}| > 3\sigma$. Therefore when calculating the local variation energy of pixel $\mathbf{p}$, the *effective neighborhood* of $\mathbf{p}$ should be within a ball centered at $\mathbf{p}$ with radius about 3σ .

B. LVEW Energy and Distance

Now the *local variation and edge-weighted energy functional* can be easily constructed by adding the total local variation energy and the total edge energy together. Precisely, for each pixel $\mathbf{p} \in \mathcal{D}$, define

$$\mathcal{E}_{(\tilde{\mathcal{G}}; \mathcal{S})}^{\text{LVEW}}(\mathbf{p}) = \mathcal{E}_{(\tilde{\mathcal{G}}; \mathcal{S})}^{\text{LV}}(\mathbf{p}) + \mathcal{E}^{\mathcal{L}}(\mathbf{p}). \quad (12)$$

For simplicity, we again omit its subscripts from now on. Naturally the total energy over $\mathcal{D}$ is

$$E^{\text{LVEW}}(\tilde{\mathcal{G}}; \mathcal{S}) = \sum_{\mathbf{p} \in \mathcal{D}} \mathcal{E}^{\text{LVEW}}(\mathbf{p}) \quad (13)$$

In Section II, we claim that if we are given a set of data and a metric defined on this data set, we can immediately construct the CVTs by the Lloyd's algorithm. On the other hand, Theorem 1 asserts that the construction of CVTs is equivalent to an energy minimization process provided the energy functional is consistent with the metric. Therefore, since we have already defined the local variation and edge-weighted energy functional, we need to find a metric which corresponds to the energy functional such that we can apply the Lloyd's algorithm to construct the LVEWCVT. We have the following proposition.

Proposition 1: Given a digital image $\mathcal{I} = \{f(\mathbf{p}) : \mathbf{p} \in \mathcal{D}\}$. Given a segmentation of $\mathcal{D}$, $\mathcal{S} = \{S_k\}_{k=1}^L$ and a set of generator functions $\tilde{\mathcal{G}} = \{\tilde{g}_k\}_{k=1}^L$. Then we define a metric between pixel $\mathbf{p}$ and the generator function $\tilde{g}_k$ at $\mathbf{p}$ as:

$$d(\mathbf{p}, \tilde{g}_k(\mathbf{p})) = \sqrt{\sum_{\mathbf{q} \in \mathbb{N}_r(\mathbf{p}) \setminus \mathbf{p}} \lambda_k H(\mathbf{p}, \mathbf{q}) |f(\mathbf{p}) - \tilde{g}_k(\mathbf{q})|^2 + 2\mu \tilde{n}_k(\mathbf{p})} \quad (14)$$

where $\tilde{n}_k(\mathbf{p})$ is the number of pixels within $\mathbb{N}_r(\mathbf{p}) \setminus (S_k \cup \mathbf{p})$. This metric is consistent with the local variation and edge-weighted energy functional defined in (13).

Proof: In order to prove that the metric defined in (14) is consistent with the energy functional defined in (13), we need to show that moving the pixel $\mathbf{p}$ from cluster S_l to cluster S_m will lead to energy decrease if $d(\mathbf{p}, \tilde{g}_l(\mathbf{p})) > d(\mathbf{p}, \tilde{g}_m(\mathbf{p}))$.

Let us rewrite (13) as

$$E^{\text{LVEW}}(\tilde{\mathcal{G}}; \mathcal{S}) = \left[\sum_{\mathbf{q} \in \mathcal{D} \setminus \mathbf{p}} \mathcal{E}^{\text{LV}}(\mathbf{q}) \right] + \mathcal{E}^{\text{LV}}(\mathbf{p}) + \left[\sum_{(\mathbf{q}) \in \mathcal{D} \setminus \mathbf{p}} \mathcal{E}^{\mathcal{L}}(\mathbf{q}) \right] + \mathcal{E}^{\mathcal{L}}(\mathbf{p}). \quad (15)$$

When we transfer a pixel $\mathbf{p} \in S_l$ to a different cluster S_m , the total energy will vary due to the four terms on the right hand side of (15). Note that during the transfer process, the generator functions are assumed to be fixed. Thus the second term in the right-hand side of (15), $\mathcal{E}^{\text{LV}}(\mathbf{p})$, stays the same.

For the variation of the first term, let us first consider the neighbor pixel $\mathbf{q} \in \mathbb{N}_r(\mathbf{p})$ of $\mathbf{p}$. Clearly, $\mathbf{p}$ is a neighbor pixel of $\mathbf{q}$ as well. Therefore, before the transfer occurs $\mathcal{E}^{\text{LV}}(\mathbf{q})$ can be written as

$$\begin{aligned} \mathcal{E}^{\text{LV}}(\mathbf{q}) &= \sum_{\mathbf{q}^* \in \mathbb{N}_r(\mathbf{q}) \setminus \mathbf{p}} \left(\lambda_{\pi_S(\mathbf{q}^*)} H(\mathbf{q}, \mathbf{q}^*) \right. \\ &\quad \times \left. |f(\mathbf{q}^*) - \tilde{g}_{\pi_S(\mathbf{q}^*)}(\mathbf{q})|^2 \right) \\ &\quad + \lambda_l H(\mathbf{q}, \mathbf{p}) |f(\mathbf{p}) - \tilde{g}_l(\mathbf{q})|^2. \end{aligned} \quad (16)$$

After we transfer the pixel $\mathbf{p}$ from S_l to S_m , the first term on the right-hand side of (16) remains the same while the second term will change, and the variation can be easily found as

$$\begin{aligned} &\lambda_m H(\mathbf{q}, \mathbf{p}) |f(\mathbf{p}) - \tilde{g}_m(\mathbf{q})|^2 \\ &\quad - \lambda_l H(\mathbf{q}, \mathbf{p}) |f(\mathbf{p}) - \tilde{g}_l(\mathbf{q})|^2 = H(\mathbf{p}, \mathbf{q}) \\ &\quad \times \{ \lambda_m |f(\mathbf{p}) - \tilde{g}_m(\mathbf{q})|^2 - \lambda_l |f(\mathbf{p}) - \tilde{g}_l(\mathbf{q})|^2 \}. \end{aligned} \quad (17)$$

Moreover it is clear that the transfer of pixel $\mathbf{p}$ from S_l to S_m only affect its neighbor pixels, therefore the total variation of the first term on the right hand of (15) can be written as

$$\sum_{\mathbf{q} \in \mathbb{N}_r(\mathbf{p}) \setminus \mathbf{p}} H(\mathbf{p}, \mathbf{q}) \{ \lambda_m |f(\mathbf{p}) - \tilde{g}_m(\mathbf{q})|^2 - \lambda_l |f(\mathbf{p}) - \tilde{g}_l(\mathbf{q})|^2 \}. \quad (18)$$

Now let us consider the edge energy variation. We first evaluate the simple case, i.e., the fourth term on the right hand side of (15). Let us denote by $n_k(\mathbf{p})$ the number of pixels within $S_k \cap \mathbb{N}_r(\mathbf{p}) \setminus \mathbf{p}$. Before moving the pixel $\mathbf{p}$, its edge energy is

$$\mathcal{E}^{\mathcal{L}}(\mathbf{p}) = \mu \sum_{(\mathbf{q}) \in \mathbb{N}_r(\mathbf{p})} \mathcal{X}_{\mathbf{p}}(\mathbf{q}) = \mu \sum_{k \in \{1, \dots, L\} \setminus \{l\}} n_k(\mathbf{p}). \quad (19)$$

Similarly, after we transfer $\mathbf{p}$ to cluster S_m , its edge energy changes to

$$\mathcal{E}^{\mathcal{L}}(\mathbf{p}) = \mu \sum_{(\mathbf{q}) \in \mathbb{N}_w(\mathbf{p})} \mathcal{X}_{\mathbf{p}}(\mathbf{q}) = \mu \sum_{k \in \{1, \dots, L\} \setminus \{m\}} n_k(\mathbf{p}). \quad (20)$$

Thus the change in the fourth term $\mathcal{E}^{\mathcal{L}}(\mathbf{p})$ on the right hand side of (15) is

$$\begin{aligned} &\mu \sum_{k \in \{1, \dots, L\} \setminus \{m\}} n_k(\mathbf{p}) - \mu \sum_{k \in \{1, \dots, L\} \setminus \{l\}} n_k(\mathbf{p}) \\ &\quad = \mu [n_l(\mathbf{p}) - n_m(\mathbf{p})]. \end{aligned} \quad (21)$$

In order to evaluate the variation of the third term on the right hand side of (15), let us first consider the energy change for each single pixel $\mathbf{q} \in \mathbb{N}_r(\mathbf{p})$. Clearly, if pixel $\mathbf{q}$ is outside of $\mathbb{N}_r(\mathbf{p})$, the transfer of $\mathbf{p}$ will not affect its edge energy. Consider another case $\mathbf{q} \in \mathbb{N}_r(\mathbf{p}) \cap S_k$. If $k \neq l, m$, obviously the edge energy of $\mathbf{q}$ will not be affected by the transferring of $\mathbf{p}$ from S_l to S_m . If $k = l$, move $\mathbf{p}$ out of cluster S_l will increase the edge energy by μ . If $k = m$, move $\mathbf{p}$ into the cluster S_m will decrease the edge energy by μ . Therefore, due to the transfer of $\mathbf{p}$ from S_l to S_m , the variation of $\sum_{\mathbf{q} \in \mathcal{D} \setminus \mathbf{p}} \mathcal{E}^{\mathcal{L}}(\mathbf{q})$ is

$$\mu n_l(\mathbf{p}) - \mu n_m(\mathbf{p}) = \mu [n_l(\mathbf{p}) - n_m(\mathbf{p})]. \quad (22)$$

Combining (18), (21) and (22) together, we then obtain the variation of the total local variation and edge-weighted energy $E^{\text{LVEW}}(\tilde{\mathcal{G}}; \mathcal{S})$ due to the transfer of the pixel $\mathbf{p}$ from S_l to S_m as follows:

$$\begin{aligned} &\sum_{\mathbf{q} \in \mathbb{N}_r(\mathbf{p}) \setminus \mathbf{p}} H(\mathbf{p}, \mathbf{q}) \{ \lambda_m |f(\mathbf{p}) - \tilde{g}_m(\mathbf{q})|^2 \\ &\quad - \lambda_l |f(\mathbf{p}) - \tilde{g}_l(\mathbf{q})|^2 \} + 2\mu (n_l(\mathbf{p}) - n_m(\mathbf{p})) \end{aligned} \quad (23)$$

which can be rewritten in a more symmetrical way as

$$\begin{aligned} &\left\{ \sum_{\mathbf{q} \in \mathbb{N}_r(\mathbf{p}) \setminus \mathbf{p}} H(\mathbf{p}, \mathbf{q}) \lambda_m |f(\mathbf{p}) - \tilde{g}_m(\mathbf{q})|^2 - 2\mu n_m(\mathbf{p}) \right\} \\ &\quad - \left\{ \sum_{\mathbf{q} \in \mathbb{N}_r(\mathbf{p}) \setminus \mathbf{p}} H(\mathbf{p}, \mathbf{q}) \lambda_l |f(\mathbf{p}) - \tilde{g}_l(\mathbf{q})|^2 - 2\mu n_l(\mathbf{p}) \right\} \\ &= \left\{ \sum_{\mathbf{q} \in \mathbb{N}_r(\mathbf{p}) \setminus \mathbf{p}} H(\mathbf{p}, \mathbf{q}) \lambda_m |f(\mathbf{p}) - \tilde{g}_m(\mathbf{q})|^2 + 2\mu \tilde{n}_m(\mathbf{p}) \right\} \\ &\quad - \left\{ \sum_{\mathbf{q} \in \mathbb{N}_r(\mathbf{p}) \setminus \mathbf{p}} H(\mathbf{p}, \mathbf{q}) \lambda_l |f(\mathbf{p}) - \tilde{g}_l(\mathbf{q})|^2 + 2\mu \tilde{n}_l(\mathbf{p}) \right\} \end{aligned} \quad (24)$$

where $\tilde{n}_m(\mathbf{p}) = |\mathbb{N}_r(\mathbf{p})| - n_m(\mathbf{p}) - 1$, is the number of pixels within $\mathbb{N}_r(\mathbf{p}) \setminus (S_m \cup \{\mathbf{p}\})$. Clearly (24) is equivalent to say the energy variation due to pixel transfer of $\mathbf{p}$ from S_l to S_m is

$$d^2(\mathbf{p}, \tilde{g}_m(\mathbf{p})) - d^2(\mathbf{p}, \tilde{g}_l(\mathbf{p})). \quad (25)$$

Therefore, we prove that the metric we defined in Proposition 1 is consistent with the energy functional (13). ■

Remark: Note that the “metric” defined in Proposition 1 beautifully combines the information of the color space and the physical space. However, the “metric” is only defined between pixels and generator functions. Further, the “metric” is not a real metric, i.e., it does not satisfy triangle inequality.

C. Local Variation and Edge-Weighted Centroidal Voronoi Tessellation: Model and Algorithms

We first define the *local Variation and edge-weighted voronoi regions* as follows.

Definition 4: Given a set of generator functions $\tilde{\mathcal{G}} = \{\tilde{g}_k\}_{k=1}^L$ defined over $\mathcal{D}$, the local Variation and edge-weighted voronoi regions are given by

$$\tilde{S}_k = \{\mathbf{p} \in \mathcal{D} : d(\mathbf{p}, \tilde{g}_k(\mathbf{p})) \leq d(\mathbf{p}, \tilde{g}_l(\mathbf{p})), \\ \text{for } l = 1, \dots, L, l \neq k\}, \\ k = 1, \dots, L. \quad (26)$$

Therefore the local variation and edge-weighted Voronoi tessellation (LVEWVT) can be easily constructed by the following algorithm.

Algorithm LVEWVT: Given a set of generator functions $\tilde{\mathcal{G}} = \{\tilde{g}_k\}_{k=1}^L$. Choose an arbitrary partition $\{S_k\}_{k=1}^L$ of $\mathcal{D}$

- 1) For every $\mathbf{p} \in \mathcal{D}$,
 - a) evaluate $d(\mathbf{p}, \tilde{g}_k(\mathbf{p}))$ for all $k = 1, \dots, L$;
 - b) move $\mathbf{p}$ to the cluster S_m such that $d(\mathbf{p}, \tilde{g}_m(\mathbf{p})) = \min\{d(\mathbf{p}, \tilde{g}_k(\mathbf{p})), k = 1, \dots, L\}$.
 - 2) If no pixel transfer occurs, return $\{S_k\}_{k=1}^L$ as $\{\tilde{S}_k\}_{k=1}^L$ and exit the loop; otherwise, go to Step 1.
-

Similar to the definition of CVT, we have the following definition:

Definition 5: For a given local variation and edge-weighted Voronoi tessellation $(\tilde{\mathcal{G}}; \tilde{\mathcal{S}})$ of $\mathcal{D}$, we call it a *local variation and edge-weighted centroidal Voronoi tessellation* (LVEWCVT) of $\mathcal{D}$ if and only if its generator functions are also the corresponding *local biased centroids* defined in (9), i.e., $\tilde{g}_k(\mathbf{p}) = \tilde{g}_k^{\text{LBC}}(\mathbf{p})$ for any $\mathbf{p} \in \mathcal{D}$, $k = 1, \dots, L$.

Moreover, we have the following theorem.

Theorem 2: The local variation and edge-weighted energy $E^{\text{LVEW}}(\tilde{\mathcal{G}}; \mathcal{S})$ is minimized only if $(\tilde{\mathcal{G}}; \mathcal{S})$ form a LVEWCVT of $\mathcal{D}$.

The following Lloyd-type algorithm can be used to construct LVEWCVTs of an image. We provide two versions, one is the basic version and the other is the accelerated version. The only difference in the accelerated version is we update the local biased centroids as soon as pixel transfer occurs. First comes the basic version.

Algorithm LVEWCVT-A: Given an integer L and choose an arbitrarily partition $\{S_k\}_{k=1}^L$ of the digital image $\mathcal{S} = \{f(\mathbf{p})\}_{\mathbf{p} \in \mathcal{D}}$

- 1) For each cluster $S_k, k = 1, \dots, L$, determine its local biased centroids $\tilde{g}_k$ by (9).
- 2) Take $\{\tilde{g}_k\}_{k=1}^L$ as the generator functions, determine the corresponding LVEWVT $\tilde{\mathcal{S}}$ according to Algorithm LVEWVT.

- 3) If the LVEWVT $\{\tilde{S}_k\}_{k=1}^L$ are the same with $\{S_k\}_{k=1}^L$, return $(\{\tilde{g}_k\}_{k=1}^L; \{S_k\}_{k=1}^L)$ and exit the loop; otherwise, set $S_k = \tilde{S}_k$ for $k = 1, \dots, L$ and go to Step 1.
-

An accelerated and improved version is given in the following.

Algorithm LVEWCVT-B: Given an integer L and choose arbitrarily a partition $\{S_k\}_{k=1}^L$ of digital image $\mathcal{S} = \{f(\mathbf{p})\}_{\mathbf{p} \in \mathcal{D}}$. Determine the *local biased centroids* $\{\tilde{g}_k\}_{k=1}^L$ and take them as the generator functions.

- 1) For every $\mathbf{p} \in \mathcal{D}$,
 - a) evaluate $d(\mathbf{p}, \tilde{g}_k(\mathbf{p}))$ for all $k = 1, \dots, L$;
 - b) suppose $\mathbf{p} \in S_l$, move $\mathbf{p}$ to the cluster S_m such that $d(\mathbf{p}, \tilde{g}_m(\mathbf{p})) = \min\{d(\mathbf{p}, \tilde{g}_k(\mathbf{p})), k = 1, \dots, L\}$.
 - c) update $\{\tilde{g}_k\}_{k=1}^L$ by (9) for $\mathbf{p}$ and its neighbor pixels;
 - 2) if no pixel is moved, return $(\{\tilde{g}_k\}_{k=1}^L; \{S_k\}_{k=1}^L)$ and exit the loop; otherwise, go to Step 1.
-

The complexity of Algorithm LVEWCVT-B is $\mathcal{O}(k \times r^2 \times L \times |\mathcal{D}|)$ where k denotes the number of iterations and $|\mathcal{D}|$ represents the size of the image. Note that, in most practical cases, we do not need to wait until the condition to exit the loops is strictly satisfied. In fact, we can calculate and record the energy E^{LVEW} of the current LVEWVT configuration at each iteration. If the decrease of the energy is within some prescribed tolerance, the algorithm will be terminated. In our numerical experiments, we stop the iteration [12] if

$$\frac{|E_{i+1}^{\text{LVEW}} - E_i^{\text{LVEW}}|}{E_i^{\text{LVEW}}} < L\%. \quad (27)$$

Adjusting the prescribed tolerance of the energy may result in slightly different final LVEWCVT configuration. All of the discussions about the convergence, stability and possible improvement of classic CVTs are applicable to the LVEWCVT without substantial changes. For detailed information, please refer to [12], [24], [30], [45].

IV. EXPERIMENTS AND DISCUSSIONS

We apply our LVEWCVT model and corresponding segmentation algorithms to synthetic and real images from different modalities. Due to its efficiency, Algorithm LVEWCVT-B will be used for all numerical experiments.

A. Guidelines for Determining Parameters

In most of the cases, it is difficult to obtain the “optimal” parameter values, but we can provide reasonable good approximations. In our proposed LVEWCVT model, the full set of parameters is listed as follows. L : the number of predefined clusters; $\{\lambda_i\}_{i=1}^L$: the weights to calculate the local intensity variation energy; μ : the weight for the edge energy; r : the size of the neighborhood of each pixel and σ which is used to calculate the weighted distance between pixels by (11).

1) *Number of Clusters L :* Both of algorithms LVEWCVT-A and LVEWCVT-B require the user to predefine the number of

clusters L . Theoretically, L can be any positive integer. However, a meaningful selection of L should be based on the number of foreground objects the user wants to extract. To identify n_o objects, we will choose $L = n_o + 1$ since the background should be accounted as another cluster. We also would like to note that fewer number of clusters may also work well since in the LVEWCVT model huge intensity variation is permissible for one cluster.

2) *Weighting Factor for Each Cluster* $\{\lambda_k\}_{k=1}^L$: Similar with the selection of L , algorithms LVEWCVT-A and LVEWCVT-B need the user to provide an initial segmentation as starting point. For the discussion of initial segmentation, please refer to Section IV-B. Suppose we have chosen the initial segmentation that has L segments. The first $L - 1$ segments are corresponding to the objects and the L th segment refers to the background. We always set $\lambda_L = 1$. For each segment k ($0 \leq k \leq L - 1$), if it is labeled as an “inhomogeneous” cluster, e.g., the k th object is complex textured or with sharp intensity/color variation, we set $\lambda_k = \lambda_{in} < 1$ for some constant λ_{in} . Otherwise, if the k th object is not with very high inhomogeneity, it can be labeled as a “homogeneous” cluster and we set $\lambda_k = \lambda_{in} = 1$. In general, by considering the energy functional of LVEWCVT, smaller λ_k implies higher tolerance of inhomogeneity. In fact, for simplicity, the user can simply regard all of the objects as “inhomogeneous” clusters.

3) *Weighting Factor of Edge Energy* μ : The LVEW energy consists of two parts, the local intensity variation $E^{LV}(\tilde{\mathcal{G}}; \mathcal{S})$ and the edge energy $E^{\mathcal{L}}(\mathcal{S})$. Therefore we need the weight μ to balance these two parts [45]. Practically, our numerical experiments indicate that we can obtain a reasonably good μ by the following procedure. Remember that we can calculate the edge energy and the local variation energy as long as we have a segmentation. Therefore, after an initial segmentation is given, we calculate a value for μ (included in $E^{\mathcal{L}}(\mathcal{S})$) such that

$$\frac{E^{LV}(\tilde{\mathcal{G}}; \mathcal{S})}{E^{\mathcal{L}}(\mathcal{S})} = M \quad (28)$$

where M is a predetermined positive number between 1 and 10. We then run the LVEWCVT algorithm one iteration and re-calculate μ using the above way, and then simply fix it and continue the remaining iterations till convergence. In all our examples, μ will be set using this automatical way except specially stated.

4) *The Parameters σ and r* : As explained in Section III-A, when $|\mathbf{p} - \mathbf{q}| > 3\sigma$, the Gaussian kernel $H_\sigma(\mathbf{p}, \mathbf{q})$ would be very close to zero. In all our examples, we set $\sigma = 3$. Since $H_\sigma(\mathbf{p}, \mathbf{q})$ is small enough for $|\mathbf{p} - \mathbf{q}| > 2\sigma$, we choose $r = 6$.

Remark: From the above discussions, there are two parameters λ_{in} and M in our LVEWCVT method that need to be well chosen. The experimental parameter values ($\lambda_{in} = 0.5, M = 4$) were applied to almost all examples in our experiments except specially stated (see Section IV-E for detailed discussion on this setting).

B. Initial Configuration—Efficiency and Sensitivity

In this section we discuss how to choose an initial configuration for the LVEWCVT method. As discussed in [45], two schemes can be used to provide the initial segmentations. The

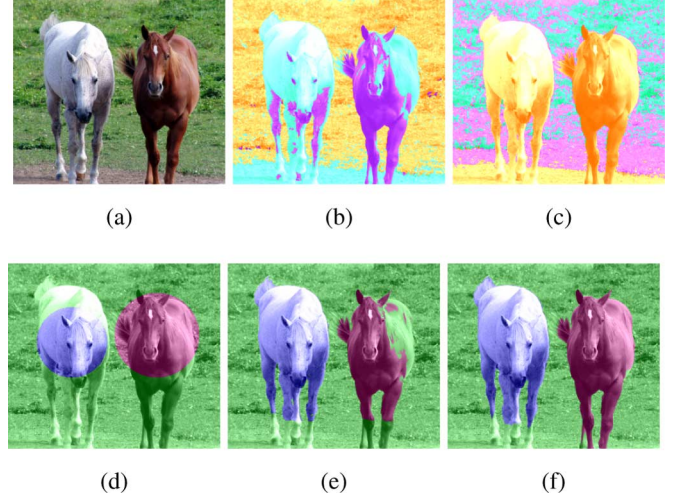


Fig. 2. Extraction of two horses by the LVEWCVT method with three clusters. (a) Original image. (b)(c) Segmentations by the classic CVT method initialized by random sampling. (d) Initial clusters by circles. (e) Segmentation by the classic CVT method with (d) as initial configuration. (f) Segmentation result by the LVEWCVT method with segments in (e) as the initial configuration.

first one is to use a classic CVT clustering and the second one is to use circles loosely labeling the objects.

Notice that the classic CVT method needs initialization as well. The most commonly used method is by random sampling [12]. However, this approach is not stable. Especially for images with rich colors, the classic CVT method with initial configurations (three clusters) by random sampling may result in completely different segmentations, e.g., Fig. 2(b) and (c). The reason is that the classic CVT method often leads to local minimizers of the energy functional and is sensitive to initial configurations.

Loosely labeling the objects by circles is clearly more stable to provide a set of initial segments. However, our numerical experiments indicate that this approach normally need too many iterations in the LVEWCVT method since the initial configuration is often far different from the final segmentation result. By using the segments presented by Fig. 2(d) as the initial guess (three clusters), our LVEWCVT method needs 67 steps to converge to that present in Fig. 2(f).

Therefore, we use the following approach to generate the initial segments for the LVEWCVT method:

- 1) We first use circles to loosely label the objects;
- 2) Apply the classic CVT method (Algorithm CVT-B [45]) using the circles as the initial segments.
- 3) Then take the segments computed from step (2) as the initial partition (clusters) for the LVEWCVT method.

We call this strategy “Initialization by CVT” (**IbCVT**). The whole process of IbCVT is illustrated by Fig. 2(d)–(f). Using the IbCVT for initialization (49 iterations used in the classic CVT method), only 17 iterations are needed for the LVEWCVT method to converge. Remember that each iteration of the classic CVT method is much less expensive than that of the LVEWCVT method, thus the IbCVT initialization approach could save a lot of running time.

Our experiments show that the IbCVT initialization almost always resulted in more stable results and much less iterations for

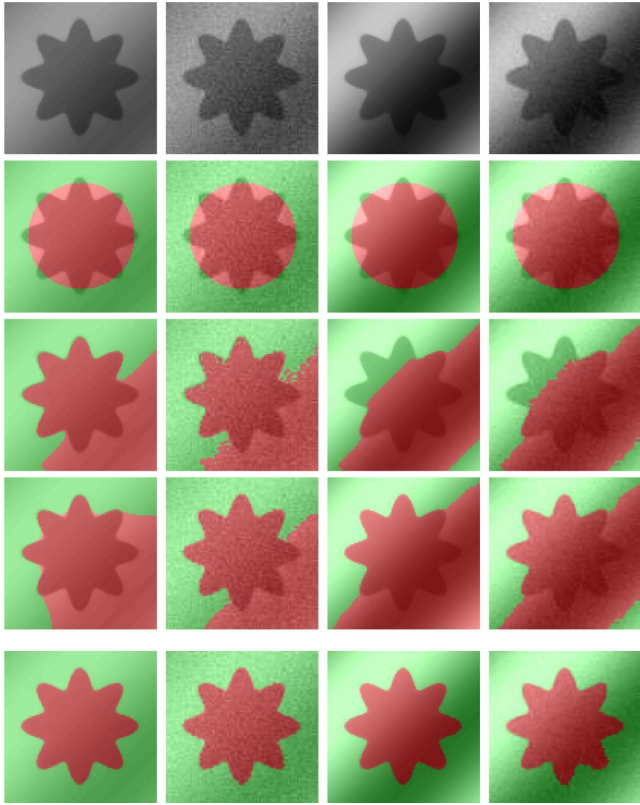


Fig. 3. Detection of a star-shaped object from inhomogeneous background by the LVEWCVT method with two clusters. Each column shows segmentation results of the top image by various algorithms. (top row) Original image. (second row) Initial clusters. (third row) Segmentations by the classic CVT method with the second row images (circles) as the initial configurations. (fourth row) Segmentations by LVEWCVT with the third row images as the initial configurations (the IbCVT initialization). (bottom row) Segmentation results by LVEWCVT with the second row images (circles) as the initial configurations.

the LVEWCVT algorithms than other initialization processes. Such initialization process was also discussed and taken in [45]. Under some “*extreme*” situations, e.g., both of the objects and background are with heavy and “*strongly correlated*” inhomogeneity, we found that the IbCVT process may not be applicable. Figs. 3 and 4 serve as two examples to illustrate this difficulty. For such extreme cases, we just need take the circles as the initial segments and then apply the LVEWCVT iteration.

In Fig. 3, all of the original images in the first row have a star-shaped object but with different degrees of intensity inhomogeneity or noise levels. The first and third images both are noiseless, but the intensity inhomogeneity of the latter one is much higher than the former. By adding Gaussian noise with standard deviations of 8.0 and 5.0 to the first and third images, we get the second and fourth images respectively. The second and third rows show the IbCVT process for each of the four images. We used two clusters, $L = 2$. The fourth row indicates that the objects extracted by the LVEWCVT method with the IbCVT initialization are overwhelmed by the background. By carefully examining the original images, we can see that both of the objects and background are with heavy and “*strongly correlated*” inhomogeneity. In other words, the intensity variations of both objects and background are in a very similar pattern, which is roughly periodic in this case. Therefore too much information of the objects and background messed up during the

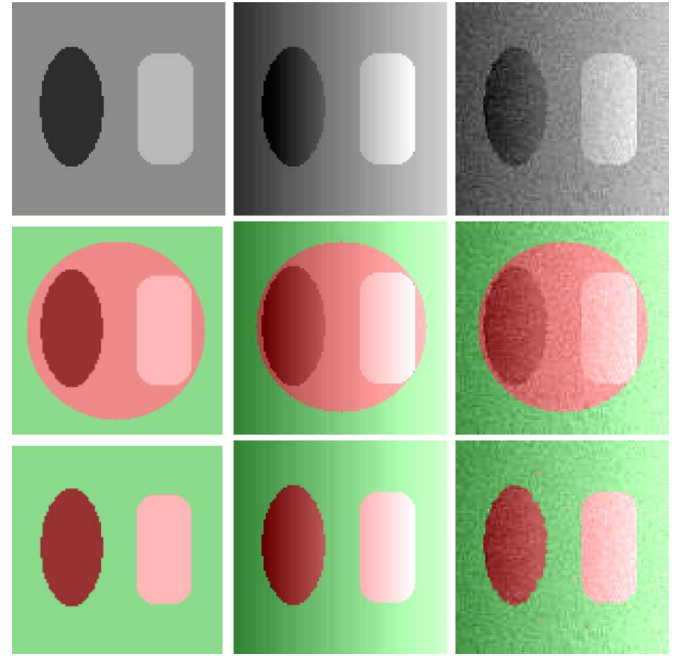


Fig. 4. Detection of two geometric objects from synthetic images with inhomogeneous background by the LVEWCVT method with two clusters. (top row) Original images. (middle row) Initial clusters. (bottom row) Final segmentation results.

IbCVT process. Instead of using the IbCVT initialization, we could start with the circles shown in the second row directly. The segmentation results of the LVEWCVT method with the second row images as the initial configurations are presented in the bottom row. We can see that, despite of the heavy artificial noise and strongly correlated intensity inhomogeneity, the LVEWCVT method accurately captured the objects. Notice that the classic CVTs can never correctly detect the star-shape objects, see the third row images of Fig. 3, since each segment has to be roughly the same in colors with respect to its centroid [45].

Fig. 4 is another example similar with Fig. 3. The difference is that in Fig. 4 the original images (the first row images) have two objects, i.e., one dark ellipse and another bright round-corner rectangle. We still set $L = 2$ although there are two objects in the foreground. For the last two images in the first row, since the inhomogeneity of the objects and background are strongly correlated, we directly start from circles presented in the second row as well. The LVEWCVT method with two clusters offered very satisfying results. Notice that both of the two objects belong to the same cluster. In the energy functional of the LVEWCVT model, the local intensity variation energy of each pixel only involves the intensities of a small neighborhood within its cluster and hence huge intensity variation is permissible for one cluster. This property of the LVEWCVT model results in a relatively low labeling cost, please refer to Fig. 12(c) (The classic CVT method can offer very good result for the first image as well since both the objects and background are uniform in colors, but it needs three clusters.)

Figs. 3 and 4 serve as two extreme examples in which IbCVT is not applicable. Fortunately, for most of the images from the real scenes, the intensities/colors of objects and the background are either heavy inhomogeneity but not strongly correlated

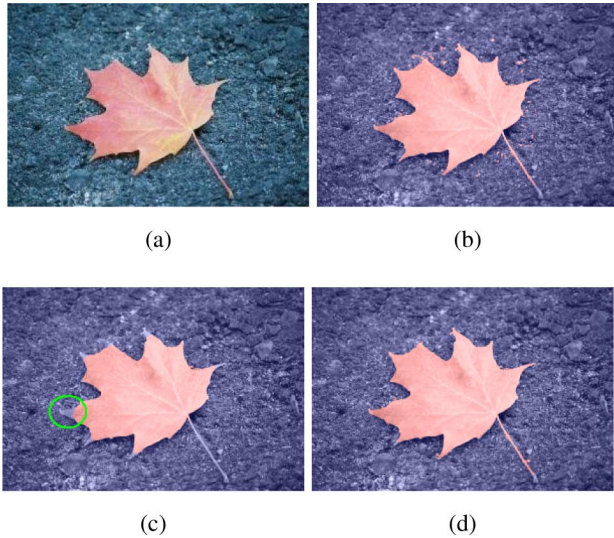


Fig. 5. Extraction of a leaf by the LVEWCVT method with two clusters. (a) Original image. (b) Initial clusters generated by the IbCVT process. (c) Segmentation results with $\mu = 10000$. (d) Segmentation results with $\mu = 1000$.

or strongly correlated but with relatively low inhomogeneity. Therefore, in most situations, the IbCVT initialization is a robust and efficient approach to initialize the LVEWCVT iteration. It is worth noting that for all examples presented below including all images from the benchmark database, we will use the IbCVT process to compute the initial segments for the LVEWCVT method.

C. Effects of Some Parameters

Next we use Fig. 5 to illustrate the effect of μ . The original image in Fig. 5 is a leaf which is roughly uniform colored. Fig. 5(b) presents the initial segments of two clusters by the IbCVT process. It is easy to find there are quite a few small fragments which in fact do not belong to the leaf. Fig. 5(c) and (d) represent final segmentation results by the LVEWCVT method using different μ . We can see that larger μ resulted in smoother boundaries and the wrongly identified small fragments were all eliminated. However, at the same time we also lost a small portion of the leaf as highlighted by the green circle in Fig. 5(c). By using smaller μ the leaf was preserved very nicely [see Fig. 5(d)], even the slender leaf stalk is preserved almost completely, but we didn't eliminate small fragments completely. For more discussions on the effect of μ , we refer to [45].

Fig. 6 is presented to illustrate the effect of λ_{in} . The original image is given in Fig. 6(a). The blue part of the wall and the "four leaved rose" are the objects we want to detect. Fig. 6(b) and (c) represent the segmentation results by the LVEWCVT method with three clusters using $\lambda_{in} = 1.0, 0.5$ respectively. With the larger $\lambda_{in} = 1.0$, Fig. 6(b) shows that objects detected by the LVEWCVT method are roughly uniform colored. The "four leaved rose" cluster is not a complete "four leaved rose" but a contour. In contrast, using a smaller $\lambda_{in} = 0.5$ resulted in a complete "four leaved rose" as shown in Fig. 6(c). Instead of a roughly uniform colored cluster, the complete rose is having sharp variation of colors, i.e., red contour and white leaves. Recall that in Section IV-A2 we state that smaller λ_{in}

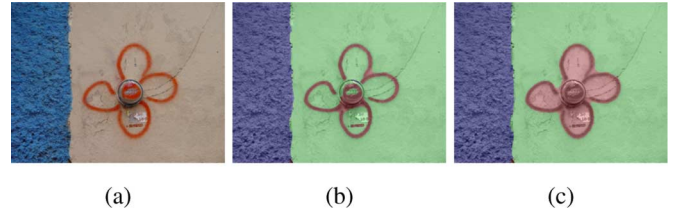


Fig. 6. Segmentation of an image by the LVEWCVT method into three clusters. (a) Original image. (b) Segmentation results by LVEWCVT with $\lambda_{in} = 1.0$. (c) Segmentation results by LVEWCVT with $\lambda_{in} = 0.5$. Objects are referred to the blue part of the wall and the "four leaved rose".

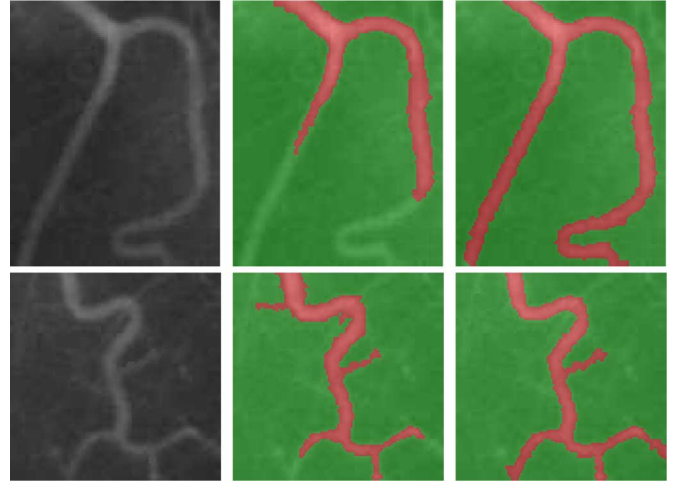


Fig. 7. Detection of blood vessels from two real CT images by the LVEWCVT method with two clusters. (left) Original images. (middle) Initial clusters by the IbCVT process. (right) Final segmentation results.

implies higher tolerance of inhomogeneity. This is exactly the case. For all of the following examples, we will always use the default parameter settings discussed in Section IV-A.

D. More Experimental Results

In this section, we will provide extensive examples to demonstrate ability of the LVEWCVT model to handle a variety of different segments types: medical images (Fig. 7), roughly uniform colored objects (Fig. 8), segments with sharp variation of colors (Fig. 9), complex textured images (Fig. 10) and multi-object images (Fig. 11).

In Fig. 7, the original images are two CT images with blood vessels. We can see that the intensity of the vessels is very close to that of the background. Even worse, for pixels far away from the vessel, the intensities also differ a lot. We applied the LVEWCVT method with two clusters and the IbCVT initialization. Clearly the classic CVT model resulted in loss of a large portion of the vessels. As expected, the LVEWCVT model greatly improves the segmentation results.

Fig. 8 shows the segmentation results by the LVEWCVT method of two uniform colored images. For the horse image, all of the fragments which do not belong to the horses disappear. Similarly, a small portion highlighted by a blue circle in the flower image was misgrouped by the classic CVT algorithm. Due to the edge energy, LVEWCVT successfully cut this portion off the true flower segment.

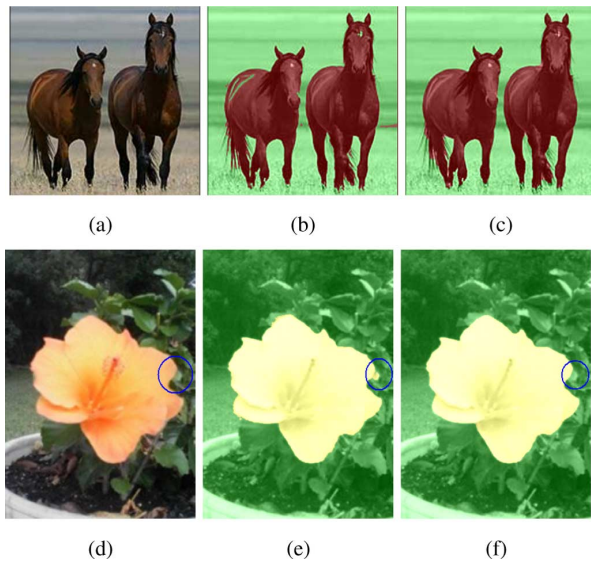


Fig. 8. Segmentation of two roughly uniform colored images by the LVEWCVT method with two clusters. (a)(d) Original images. (b)(e) Initial clusters by the IbCVT process. (c)(f) Final segmentation results.

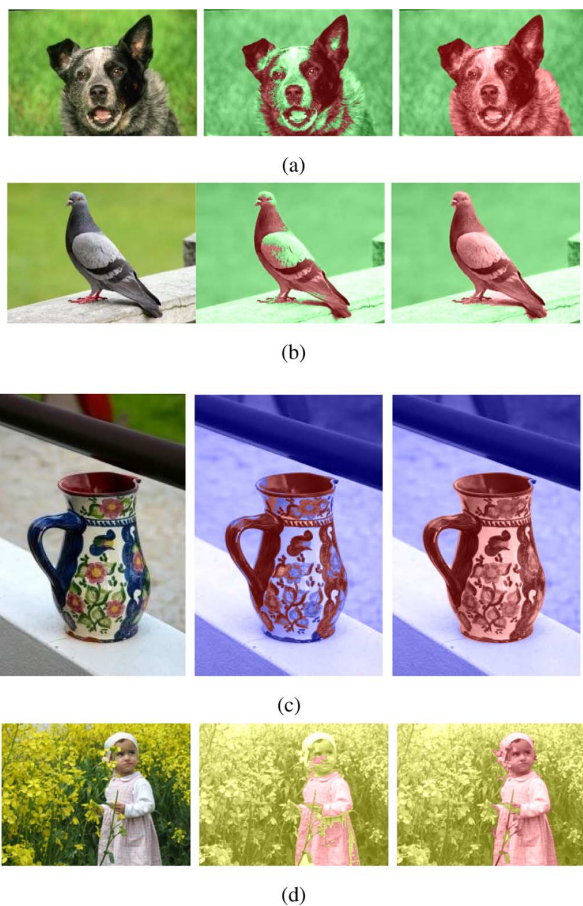


Fig. 9. Segmentation of images with sharp intensity variations by the LVEWCVT method with two clusters. (left column) Original images. (middle column) Initial clusters by the IbCVT process. (right column) Final segmentation results.

Applications of LVEWCVT to some complex images with quite sharp intensity variation are given in Fig. 9. It is easy to see that, the objects (i.e., the dog, the pigeon, the pot and the girl)

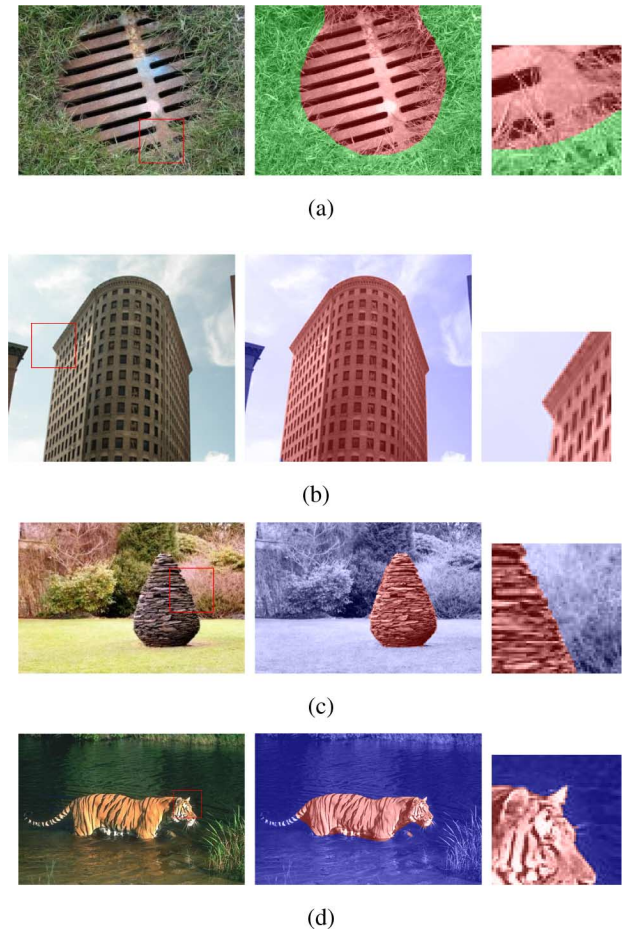


Fig. 10. Segmentation of four texture images by the LVEWCVT method with two clusters. (left column) Original images. (middle column) Final segmentation results. (right column) Zoomed-ins of the segmented images.

are not uniformly colored in these cases. The color of the dog's eye area varies sharply within its forehead area. Obviously, there are also huge difference between the dog's tone and its face in color. Similarly, the color of the pigeon's back differs a lot with its head, not mention its red feet. For the pot, it is very complicatedly colored. Fig. 9(d) is a more complex example. The little girl has black hair and black eyes with a white hat and a pink skirt, standing in yellow flowers. Because of the sharp intensity variation, it is no surprise that the initialization given by the IbCVT process are not as good as those in Figs. 5 and 8. Again, the LVEWCVT method provided us pretty accurate segmentations and thus the LVEWCVT method is in general robust to the initialization segments generated by the IbCVT process.

Texture images are very common in natural scenes. In Fig. 10, we present segmentation results by the LVEWCVT method of four different type of texture images. Two clusters were used for all images. Fig. 10(a) is a manhole cover with sharp variation of colors. Fig. 10(b) is a building with high contrast. Fig. 10(c) is a complicated texture image. The object in Fig. 10(c) consists of many irregular stone blocks and is in a complex background. Moreover, the tiger in Fig. 10(d) has orange fur with black stripes. The last column of Fig. 10 is the corresponding zoomed portion of the original images

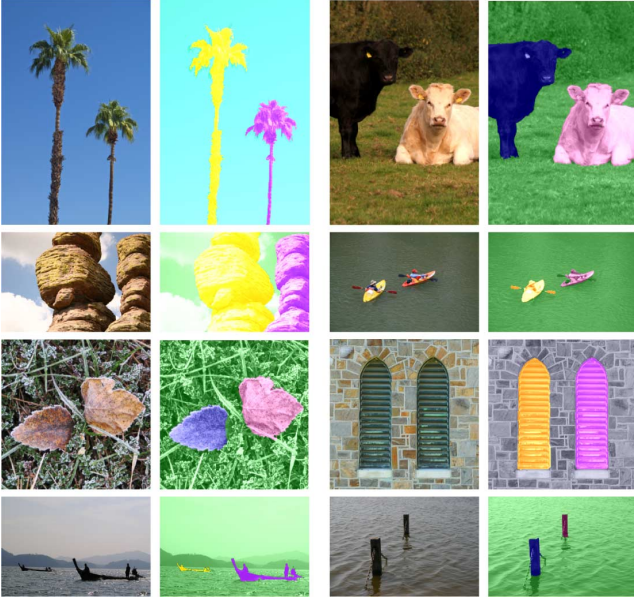


Fig. 11. Segmentation results over various images by the LVEWCVT method with three clusters. (first and third columns) Original images. (second and fourth columns) Segmentation results. All of the images are from the two objects Weizmann database.

highlighted by a red square. Again, the LVEWCVT method was able to successfully generate accurate segmentations for all these images.

As we said at the beginning of this section, there is no essential difference for the LVEWCVT method to handle single or multiple objects images (just use more clusters). Fig. 11 shows several typical segmentation examples by the LVEWCVT method for two objects images. The first three rows present segmentation results of roughly uniform colored images, images containing objects with sharp intensity/color variations and complex texture images. For images in the fourth row, we need to separate the objects with their reflections in water. We can see that the segmentation results by the LVEWCVT method are still satisfactory.

E. Quantitative Comparisons of the LVEWCVT Method With Other State-of-Art Algorithms

It is often quantitatively difficult to say how good a segmentation is. Different users may want completely different segmentation results according to various purpose. In order to provide the community some standards or benchmarks for evaluation and comparison of segmentation algorithms, there have been several important attempts to set up evaluation databases proposed in [1], [27], and [34]. In this paper, we quantitatively evaluate our algorithm using the Weizmann databases [1].

Total seven methods will be compared with each other: LVEWCVT, CVT [12], EWCVT [45], segmentation by probabilistic Bottom-Up aggregation and cue integration (we denote it as SPA for convenience) [1], segmentation by weighted aggregation (SWA) [38], normalized cuts [39] and Mean Shift [8]. We note that data and images of the last four algorithms (SPA, SWA, Ncuts and Mean Shift) we used in Figs. 12–14 are from

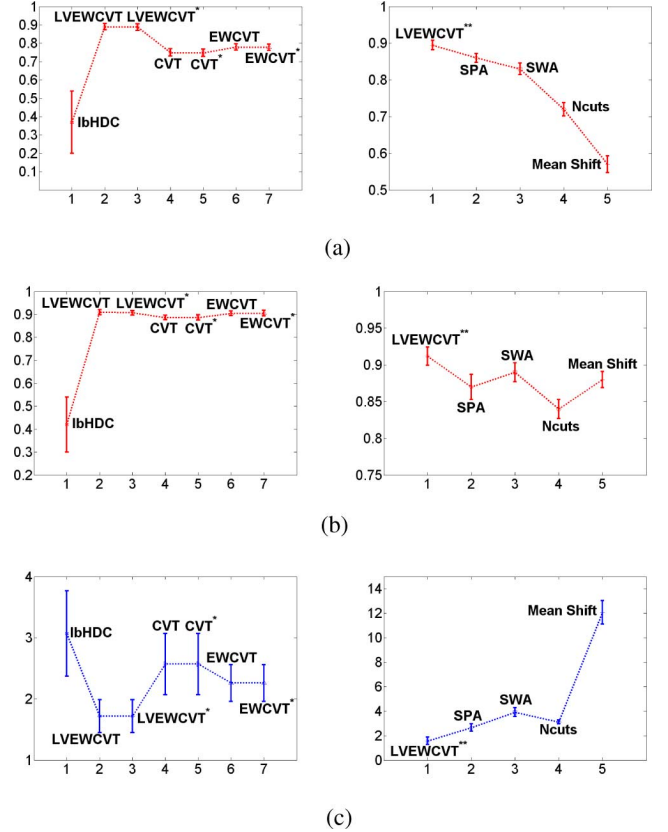


Fig. 12. Quantitative comparisons of segmentation accuracy and labelling cost using the single object Weizmann database. The figures in the left and right columns are for **E1** and **E2**, respectively. (a) F-measure scores for single-segment coverage test. (b) F-measure scores for multi-segment coverage test. (c) Average number of fragments to cover the foreground objects.

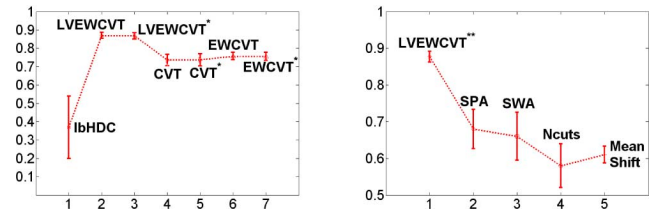


Fig. 13. Comparison of F-measure scores on the two objects Weizmann database. The results are based on single segment coverage test. The left and right figures are for **E1** and **E2**, respectively.

[1] and http://www.wisdom.weizmann.ac.il/~vision/Seg_Evaluation_DB/scores.html.¹

As stated in Section IV-A, there are two parameters λ_{in} and M that need to be well chosen for the LVEWCVT method. Let $\mathcal{A} = \{0.25, 0.5, 0.75, 1.0\}$ and $\mathcal{B} = \{1, 2, \dots, 10\}$. We select (λ_{in}, M) from the set $\mathcal{A} \times \mathcal{B}$ such that the LVEWCVT method yields the highest F-measure score over the entire databases (31). By the above strategy, we found that $(\lambda_{in} = 0.5, M = 4)$ is the optimal one. These parameters were then applied to all database examples. We also would like to

¹It was stated in [1] that the authors tested forty sets of parameters for Mean Shift and SWA respectively, and then selected the one which yielded the best F score for the entire database. For Ncuts, the only parameter is the number of clusters and the authors tested from 2 to 10, and again chose the one that produced the best F score. SPA is parameter free.



Fig. 14. Visual comparisons of some segmentation results of LVEWCVT with SPA, SWA, Ncuts and Mean Shift. The first two images are from Berkeley Benchmark. The remaining images are all from the single object Weizmann database. (a) Original image. (b) LVEWCVT. (c) SPA. (d) SWA. (e) Ncuts. (f) Mean Shift.

note that such a strategy sometimes may lead to over-fitting and that, ideally, parameters should be learned on an independent train set. Given the low number of free parameters, the results should still be comparable to other methods.

To give a fair comparison of the above algorithms, we conduct two different sets of quantitative experiments.

E1: Start with the hand-drawn circles as initial segments (**IbHDC**) for CVT, EWCVT and LVEWCVT. To distinguish with CVT, EWCVT and LVEWCVT initialized by IbCVT, we denote the three CVT based algorithms initialized by IbHDC as CVT*, EWCVT* and LVEWCVT*. To keep things simple, we use IbHDC to initialize IbCVT as well. Then we apply LVEWCVT, LVEWCVT*, CVT, CVT*, EWCVT and EWCVT* to all the benchmark images and compare the results of the 6 methods with IbHDC. This test

would qualitatively demonstrate the improvement of LVEWCVT with respect to our prior works and manual initialization.

E2: Use SPA to produce the initial segments for LVEWCVT such that LVEWCVT becomes an automatic method. We denote LVEWCVT initialized by SPA as LVEWCVT**. Then we use LVEWCVT** to segment the benchmark images and compare the results against SPA, SWA, Ncuts and Mean Shift. Therefore this test would be a fair comparison with respect to existing automatic algorithms.

By assessing consistency of the segmentation results generated by LVEWCVT, CVT, EWCVT, SPA, SWA, Ncuts and mean shift with the ground truth segmentations, we can readily provide a quantitative measurement and comparison of accuracy of these algorithms. In this paper, we use the standard F-measure

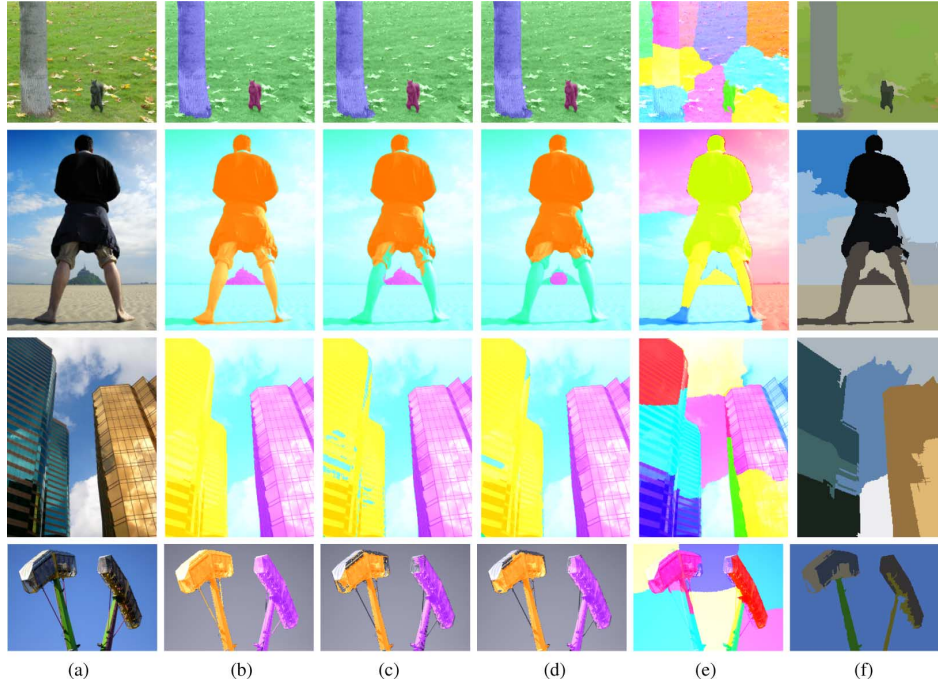


Fig. 15. Visual comparisons of some segmentation results of LVEWCVT with CVT, EWCVT, Ncuts and Mean Shift. All of the original images are from the two objects Weizmann database. (a) Original image. (b) LVEWCVT. (c) CVT. (d) EWCVT. (e) Ncuts. (f) Mean Shift.

sure score. Let us denote the precision and recall by P and R such as

$$P = \frac{\text{number of correctly predicted foreground pixel}}{\text{number of predicted foreground pixel}} \quad (29)$$

$$R = \frac{\text{number of correctly predicted foreground pixel}}{\text{number of true foreground pixel}}. \quad (30)$$

Then the F-measure is defined as the harmonic average of P and R :

$$F = \frac{2RP}{P + R}. \quad (31)$$

The labeling cost is defined as the averaged number of segments needed to cover the objects.

Similar with [1], we perform two tests over the two subsets of Weizmann databases: the single object and the two objects databases. For each test, we perform **E1** and **E2** separately. Figs. 12 and 13 show the quantitative results over these two databases respectively.

In the first test, we perform the so called “*single-segment coverage test*”. In other words, for each object, we look for a “*single*” segment which best fits that object according to the F-measure score. From Figs. 12(a) and 13, we can see that LVEWCVT and its variants give the best performance for both **E1** and **E2**. As we mentioned before, for the CVT-based algorithms, e.g., LVEWCVT, CVT, EWCVT and the corresponding variants, there is no essential difference when applying them to handle the single or multiple objects/clusters cases. Therefore,

the performance curves in the left columns (**E1**) of Figs. 12(a) and 13 are pretty close to each other. Notice that for **E1** (left column of Figs. 12 and 13), the performances of LVEWCVT, CVT and EWCVT are very close to LVEWCVT*, CVT* and EWCVT* correspondingly. As we mentioned in Section IV-B, the reason main is that IbCVT is in fact an acceleration strategy of IbHDC. In addition, for the single object Weizmann database, the right image of Fig. 12(a) shows that SPA offers the most stable performance since it has the smallest variance. The main reason for Mean Shift having poor performance is that Mean Shift is good at extracting uniform colored objects but often fails to handle textures.

The second test is the “*multi-segment coverage test*”. For each object, we seek a union of segments which fits the best the object according to the F-measure score. For the single object Weizmann database, the performances of all the algorithms are pretty close to each other. This is mainly because any complicated objects can be divided into several pieces which are relatively simple. Please refer to Fig. 12(b) for test two. Fig. 12(c) demonstrates LVEWCVT and its variants need the least number of fragments to cover the foreground objects and thus the minimum labeling cost.

Finally, we present some typical segmentation results by the above algorithms in Figs. 14 and 15. The segmentation results of Ncuts² and Mean Shift³ presented in Fig. 15 were obtained by running the packages downloaded from the authors’ websites with default parameters (particularly, for Mean Shift, the default kernel sizes $[hs; hr] = [7; 6.5]$).

²Code is from Jianbo Shi’s website http://www.cis.upenn.edu/~jshi/software/files/NcutImage_7_1.zip

³Code is from http://coewww.rutgers.edu/riul/research/code/EDISON/edison_binary.zip

V. CONCLUDING REMARKS AND FUTURE WORK

In this paper, we propose a LVEWCVT model for image segmentation, which combines the local variation information of color/intensity and the edge length. Similar to the classic CVT and the EWCVT models, the LVEWCVT model preserves the powerful capability to segment uniform colored regions/segments [45]. On the other hand, the LVEWCVT is able to handle much complicated cases with intensity inhomogeneities. Most important contributions of this new model include introduction of the weighting factors for the inhomogeneous regions and utilization of the local variations of the color/intensity into CVT-based image segmentation methods. For the classic CVT or its generalizations, the associated centroids can be viewed as a piecewise constant function over the computational domain, however, in the LVEWCVT the centroids become a vector-valued piecewise smooth function. From the practical point of view, we have shown advantages of the proposed LVEWCVT model by extensive numerical examples and quantitative evaluations. The comparisons of the LVEWCVT method with several other state-of-art algorithms further demonstrate its excellent performance. It would be very important and useful to develop schemes which can automatically adjust the value of μ and λ_{in} for different local regions/clusters of the images to further improve performance of the LVEWCVT method.

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